

ARUSHI ARORA

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RESEARCH EXPERIENCE

July 2026 - September 2026

3-Months Postdoc training programme

(Mentor: Dr. Andrea Kirsch)

Enrollment: OFM Lab, RC-FEMS (Research Center Future Energy Materials and Systems), Ruhr University, Bochum (Germany)

November 2025- January 2026

Post-Doctoral Research Scientist

(Mentor: Prof. S.K. Mehta)

Enrollment: Department of Chemistry and Center of Advanced Studies in Chemistry, Panjab University, Sector 14, Chandigarh (UT)-160014 (India)

Project: Development of electrochemical sensors for sensing OATB antibodies

EDUCATION

August 2020- August 2025

Ph.D. in Materials Science

(PhD thesis advisor: Dr. Menaka Jha)

Enrollment: Institute of Nano Science and Technology (INST) & Indian Institute of Science Education and Research (IISER), Sector 81. Mohali-140306 (India)

Thesis title: Efficient Electrochemical Hydrogen Production through Surface-Engineered Metal Sulphide Surface

September 2017- September 2019

Master of Science (M.Sc.) in Chemistry (Physical Chemistry)

(Thesis advisor: Prof. S.K. Mehta)

Enrollment: Department of Chemistry and Center of Advanced Studies in Chemistry, Panjab University, Sector 14, Chandigarh (UT)-160014 (India)

Thesis title: Magnetically retrievable Ce-doped Fe₃O₄ nanoparticles as scaffolds for the removal of azo dyes

August 2014- August 2017

Bachelor of Science (B.Sc.) in Chemistry

Enrollment: Department of Chemistry and Center of Advanced Studies in Chemistry, Panjab University, Sector 14, Chandigarh (UT)-160014 (India)

LABORATORY SKILLS & EXPERTISE

- **Hands on Training :** X-ray Photoelectron Spectroscopy (XPS) ; UV Photoelectron spectroscopy (UPES) ; X-ray Diffraction (XRD) ; Transmission electron microscopy (TEM) ; Inductive coupled plasma spectrometry (ICP-MS) ; Differential Scanning calorimetry (DSC) ; Surface Area measurement (BET) ; UV-Visible spectrophotometer ; Fourier transform infrared spectroscopy

(FTIR) ; CHNS Analyser Microwave digester and synthesizer

- **Advanced synthesis strategies** for metal sulphide and oxide nanomaterials.
- **Electrochemical Workstation** : Metrohm Autolab PGSTAT30; Origalys origaflex OGF-500
- **Water Splitting; Hydrogen evolution; Oxygen evolution ; Ethanol oxidation; Chlorine Evolution**
- **Waste to wealth conversion** - Iron extraction from scrap ; metal recovery from waste batteries
- **Environment remediation**- Waste dye and antibiotics removal from contaminated water
- **Field emission** from metal borides and sulphides
- **Renewable Energy** technologies

PUBLICATIONS

1. **Arushi Arora**, Sushma Kumari, Nausad Khan, Ritika Wadhwa and Menaka Jha, Electrochemical Surface Engineering of $\text{Co}_3\text{Ni}_3\text{S}_8$ for Ultralow Overpotential Hydrogen Evolution, *Renewable Energy*, 2026, 125795
2. **Arushi Arora**, Menaka Jha and S.K. Mehta, Upcycling of scrap Iron for synthesis of Iron sulphide and its application in hydrogen evolution, *Waste Management*, 2025, 204, 114927
3. **Arushi Arora**, Anima Mahajan, Nausad Khan, Santanu Ghosh, Menaka Jha, Design of a new process for the stabilization of $\text{FeS-Bi}_2\text{S}_3$ hybrid nanostructure and its application as a field emitter, *Nanoscale*, 2024, 16 (47), 21847-21855
4. **Arushi Arora**, Ritika Wadhwa, Krishna K. Yadav, Ankush, Menaka Jha, Enhanced electrochemical oxygen generation from sillenite phase of bismuth iron oxide ($\text{Bi}_{24}\text{Fe}_2\text{O}_{39}$) ultrafine particles stabilized at room temperature, *Journal of Electroanalytical Chemistry*, 2024, 118154
5. **Arushi Arora**, Sunaina, Ritika Wadhwa, Menaka Jha. Conversion of scrap iron into ultrafine $\alpha\text{-Fe}_2\text{O}_3$ nanorods for efficient visible light photodegradation of ciprofloxacin, *New Journal of Chemistry*, 2022, 46, 12, 5861-5868
6. **Arushi Arora**, Kritika Sood, Menaka Jha, Understanding an insight of excellent hydrogen production from cobalt molybdenum sulphide flowers assembled from ultrafine sheets, 2026 (Under revision)
7. Nausad Khan, **Arushi Arora** and Menaka Jha, Efficient hydrogen evolution from vanadium tetra sulphide microflower assembled from sheets, *Electrochimica Acta*, 2026, 573, 149434
8. Nausad Khan, Ritika Wadhwa, **Arushi Arora**, Menaka Jha, Facile solvent free process for synthesis of tungsten disulphide sheets and their application for electrochemical hydrogen generation, *Materials Chemistry and Physics*, 2026, 375, 132358
9. Manju, Swati Sharma, **Arushi Arora**, Menaka Jha, Ceria-Graphite Composite for Arsenate Removal from Groundwater : Strategic Surface Anchoring for Enhanced Performance, *New Journal of Chemistry*, 2026, 50, 3495-3508
10. Ritika Wadhwa, **Arushi Arora**, Ankush, Sushma Kumari, Menaka Jha, Enhanced photogenerated carrier assisted ethanol oxidation via surface modification with perovskite to develop pn type $\text{NiTiO}_3\text{-NiO}$ heterostructure *Materials Research Bulletin*, 2026, 195, 113865.
11. Kritika Sood, **Arushi Arora**, Aman K. Bhasin, Menaka Jha, K.K. Bhasin, Boosting electro-oxidation efficiency via chemically synthesized $\text{GdCoO}_3/\text{g-C}_3\text{N}_4$ composite : A promising anode for renewable energy systems, *Electrochimica Acta*, 2025, 542, 147552
12. Kritika Sood, **Arushi Arora**, Aman K. K. Bhasin, Kuldip K. Bhasin, and Menaka Jha, Green Synthesis of NiO Nanoparticles Using Waste Tea Extract and Their Utilization as Potential Electroanode for Ethanol Oxidation, *Energy Technology*, 2025, 2500280
13. Supriya Rana, **Arushi Arora**, Sushma Kumari, Ahmed Belal Salik Usmani, Sunaina, Surinder K. Mehta, Menaka Jha, CTAB-mediated morphology control growth of porous nickel oxide anchored on graphite for efficient urea oxidation, *Materials Research Bulletin*, 2026, 193, 113689
14. Anima Mahajan, Menaka Jha, **Arushi Arora**, GR Umapathy, Santanu Ghosh, Synthesis of $\text{MoS}_2@\text{NdS}$ heterostructures featuring augmented field emission performance, *J. of Materials chemistry A*, 2024, 12, 37, 25274-25290

15. Arushi Sharma, **Arushi Arora**, S.K. Mehta, Au Nanoparticles Decorated Graphitic Carbon Nitride Nanosheets as a Sensitive and Selective Fluorescence Probe for Fe^{3+} and Dichromate Ions in Aqueous Medium, *Chemosphere*, 2024, 363, 142834
16. Ahmed Belal Salik Usmani, Supriya Rana, **Arushi Arora**, Krishna K. Yadav, Heena Sammi, Neha Sardana, Menaka Jha, Electrochemical oxygen generation from VO_2 nanoflakes decorated onto graphite sheet, *Journal of Alloys and Compounds*, 2024, 976, 173058
17. Kritika Sood, Ritika Wadhwa, **Arushi Arora**, K.K. Bhasin, Menaka Jha, Harnessing LaCoO_3 Perovskite as a Sustainable Catalyst for Electrochemical Chlorine Evolution in Dye-Contaminated Saline Wastewater, *ACS Applied Energy Materials*, 2024, 7, 21, 9902-9910
18. Nausad Khan, Anima Mahajan, **Arushi Arora**, Kritika Sood, Sushma Jangra, Santanu Ghosh, Menaka Jha, Excellent Field Emission from Anisotropic Tungsten Trioxide Nanostructures Derived Through Environmental Friendly Green Process, *Material Chemistry and Physics*, 2024, 320, 129364
19. Gulshan Kumar, Santanu Ghosh, **Arushi Arora**, Menaka Jha, Professor Pankaj Srivastava, Effect of transition metal decoration on field emission properties of vertically aligned carbon nanotubes: An interplay between conventional parameters and occupancy of 3d and 4s state, *Journal of Materials Science: Materials in Electronics*, 2024, 35, 2071
20. KK Yadav, G Kumar, **Arushi Arora**, S Ghosh, M Jha, An insight of enhanced field emission from vertically oriented $\text{La}_x\text{Nd}_{1-x}\text{B}_6$ nanorods, *Materials Chemistry and Physics*, 2022, 279, 125694
21. Shivani Uppal, **Arushi Arora**, Sanjeev Gautam, Suman Singh, RJ Choudhary, SK Mehta, Magnetically retrievable Ce-doped Fe_3O_4 nanoparticles as scaffolds for the removal of azo dyes, *RSC Advances*, 2019, 9, 23129-2314

BOOK CHAPTERS

1. **Arushi Arora** and K.K. Yadav, Potentiometric devices for Biomarkers, *Sensing materials and devices for biomarkers*, 2024, 146-165.
2. **Arushi Arora** and Menaka Jha, Green and sustainable future and conclusion, *Industrial applications of nanoemulsions*, 2024, 267-273
3. Menaka Jha, **Arushi Arora**, Kritika Sood, Wastewater Treatment Using Nanoadsorbents Derived from Waste Materials, *Waste to Profit*, 2023, 147-164.
4. Ritika Wadhwa, **Arushi Arora** and Krishna Yadav, Advanced fiber materials in optical and photonic application, *Journal of Optical Communications*, 2023, 191-218.
5. Ritika Wadhwa, **Arushi Arora** and Krishna Yadav, Overview of advanced fiber materials, *Fiber Materials: Design, Fabrication and Applications*, 2023, 1.
6. KK Yadav, **Arushi Arora**, S Jangra, M Jha Chemically Modified Carbon Nanotubes in 3 D and 4 D Printing, *Chemically Modified Carbon Nanotubes for Commercial Applications*, 2023, 419-439.

PATENT

1. Process of preparing silica-iron oxide from iron ore slime, Santanu Sarkar, Niloy Kundu, Tamal Kanti Ghosh, Menaka Jha, Sujit Kumar Guchhait, Krishna Yadav, Sunaina, **Arushi Arora**, 2023, Patent Application Number: 202231018701
2. Method of separating iron and cobalt from their salt and solid mixture, Menaka Jha, S.K. Mehta, Supriya Rana, **Arushi Arora**, 2023, Patent Application Number: 202411001965

PRESENTATIONS AND CONFERENCE PROCEEDINGS

1. 7th Edition of hybrid international conference on "Nanotechnology for Better Living" NBL-2021, Srinagar -2021
2. 1st annual meeting of Energy and Environmental unit, INST Mohali -2021
3. 1st bilateral meeting on Smart Materials for Energy and Environmental Technology (Smart meet-2022), INST Mohali -2022
4. IITM-RSC desktop Seminar on Environmental sciences -2022
5. Bangalore India Nano (Govt. of Karnataka), Virtual event -2022
6. Frontiers in Chemical Sciences, IIT Guwahati -2022
7. 2nd CSIO-INST Bilateral Meet, INST Mohali -2023

8. 3rd Research Scholar's Day, INST Mohali -2023
9. 34th MRSI AGM and 5th Indian Materials Conclave, IIT-BHU, Varanasi -2023
10. Workshop on "Molecular simulations for Materials Design", NIT Calicut -2024
11. International conference on materials for Energy, Environment and Healthcare, NIT Calicut -2024

AWARDS

- **Gate 2020** Qualified (AIR-248), Subject- Chemical Sciences
- **Best Poster Award** at Nanotechnology for better living 2021, 7th to 11th September 2021 - NIT, Srinagar
- DST India JRF-SRF fellowship

LANGUAGE PROFICIENCY

- Hindi (Native)
- English (Fluent: Reading, writing, speaking and listening)

EXTRACURRICULAR ACTIVITIES

- Playing Table Tennis; Lawn Tennis; Reading fiction and classics

REFREES

1. Dr. Andrea Kirsch

Group leader and Junior Professor

Oxidic Functional Materials group, RC-FEMS

Fakultät für Chemie und Biochemie und dem Research Center Future Energy Materials and Systems (RC-FEMS), Ruhr University, Bochum, Germany

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2. Prof. S.K. Mehta

Professor, Department of Chemistry and CAS Coordinator,

Ex- Vice Chancellor, University of Ladakh,

Coordinator CRIKC, Local Coordinator GIAN

Ex-Honorary Director, SAIF, PU

Ex-Chairman, Dept. of Chemistry PU

Adjunct Professor, Shoolini University, Solan (H.P.)

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3. Dr. Menaka Jha

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